

Amir Mirzai Golpayegani

📍 Calgary, AB, Canada

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Summary

M.Sc. Computer Science graduate with experience building Python/C++ data systems, SQL-backed applications, data pipelines, and analytical workflows. Skilled in data preparation, modeling, validation, documentation, and visualization-oriented software, with a record of learning new cloud technologies quickly and communicating technical work clearly.

Education

M.Sc. in Computer Science <i>University of Calgary</i>	GPA: 3.94/4.0	Sep 2022 – Feb 2026 <i>Calgary, Canada</i>
B.Sc. in Computer Engineering <i>Sharif University of Technology</i>	GPA: 3.69/4.0	Sep 2016 – Feb 2021 <i>Tehran, Iran</i>

Technical Skills

Programming: Python, SQL, C++, JavaScript

Data / Analytics: NumPy, Pandas, SciPy, PostgreSQL, data preparation, data modeling, validation

Cloud / Engineering: AWS, Docker, Git, Linux, GitHub Actions, REST APIs, Django, Node.js

Visualization / Automation: Streamlit, Polyscope, OpenGL, GLSL, n8n, OpenClaw, SearXNG, Telegram API

ML / Geospatial: PyTorch, Scikit-learn, TensorFlow, Google Earth Engine API, DGGs, GDAL, Rasterio, SNAP

Work Experience

Research Software Developer Sep 2022 – Feb 2026
BigGeo / Mitacs, University of Calgary *Calgary, Canada*

- Built C++/Python software for scalable preparation, storage, analysis, and retrieval of Earth observation data in a collaborative research-industry project.
- Automated SAR and optical imagery preprocessing using GDAL, Rasterio, SNAP, and satellite data APIs.
- Developed DGGs-based data pipelines to transform satellite imagery into structured tensor datasets for analysis and retrieval.
- Improved encoding performance with C++ multithreading, reducing 10-tensor processing from 233s to 26s.
- Validated outputs and documented data-processing methods, performance results, and implementation details for thesis and publication workflows.

Back-End Developer Jun 2020 – Jul 2022
Asr Gooyesh Pardaz *Tehran, Iran*

- Developed Django REST APIs for NLP, text-to-speech, and speech-to-text service plans backed by PostgreSQL.
- Built authentication features including registration, Google login, password reset, and email/phone verification.
- Implemented backend logic for payment-service integration and storing plan/payment records in a relational database.
- Translated service requirements into database-backed features and supported implementation, debugging, and production-oriented maintenance.

Deep Learning Teaching Assistant Jul 2026
Neuromatch Academy *Remote, United States*

- Mentored an international pod of ~15 students through Neuromatch Academy's intensive three-week Deep Learning course.
- Facilitated hands-on PyTorch tutorials spanning MLPs, CNNs, attention/transformers, diffusion generative models, and reinforcement learning.
- Guided project groups through end-to-end deep learning research projects, from question formulation and literature review to model implementation and final presentations.

Teaching Assistant Dec 2022 – Apr 2025
Department of Computer Science, University of Calgary *Calgary, Canada*

- **Developing Big Data Applications:** Mentored +25 students on AWS-based data pipelines, cloud storage, debugging, and scalable data workflows.
- **Working with Data at Scale:** Supported +20 students with data preparation, pipeline design, and high-volume analysis.
- **Database Management Systems:** Assisted +75 students with SQL, relational database design, course projects, grading, and exam support.

Research and Technical Projects

Spatiotemporal Earth Observation Data System Using DGGs Sep 2022 – Feb 2026
First-author publication | *M.Sc. thesis* | C++, Python, DGGs, Wavelets, B-splines, Time-Series

- Designed a DGGS-based framework for real-time multiresolution management of satellite imagery.
- Developed a tensor-based data model for direct access to arbitrary regions at multiple resolutions.
- Implemented a triangular wavelet scheme to support multiresolution retrieval without image-pyramid redundancy.
- Built and validated reusable components for data storage, retrieval, transformation, and performance evaluation.
- Retrieved 105,489 DGGS cells per time-series query in 13.9s, compared with about 220s using global retrieval.

DEM Super-Resolution Framework

May 2024 – Feb 2026

Python, PyTorch, CNNs, Scientific Imaging, Google Earth Engine API, Polyscope, Data Processing and Visualization

- Trained deep ResNet models for 4× DEM super-resolution, reconstructing 30m elevation data from 120m inputs.
- Built a Streamlit pipeline using Rasterio to parallelize GEE DEM tile downloads and systematically prepare mountainous terrain data worldwide.
- Implemented repeatable training and evaluation workflows with statistical and terrain-aware metrics such as RMSE, slope, TRI, TPI, and wavelet-based errors.
- Developed a Polyscope-based 3D visualization tool to compare source, reconstructed, and reference terrain data from multiple angles.

Agentic Job Search Automation Platform

May 2026 – Present

OpenClaw, n8n, LLMs, AI Agents, Docker, Workflow Automation, SearXNG, Telegram API

- Building an agentic AI system to search jobs, analyze job fit, and generate tailored application materials.
- Designed LLM-based workflows for resume/job matching, skill-gap analysis, and application decision support.
- Integrated Google Sheets, Google Drive, and Telegram API to track jobs and send automated application alerts.
- Prototyped workflow automation in n8n and Docker before migrating toward a more flexible OpenClaw-based agent system.

Persian Text Normalizer for NLP

Oct 2016 – Feb 2021

Python, NLP, Text Processing

- Developed a Python-based text normalization toolkit to standardize Persian text for downstream NLP pipelines.
- Implemented preprocessing rules for cleaning, formatting, and preparing Persian text data for language-processing tasks.

Publication

Amir Mirzai Golpayegani, Mahmudul Hasan, Faramarz F. Samavati. *Real-Time Multiresolution Management of Spatiotemporal Earth Observation Data Using DGGS*. Remote Sensing, 2025. [doi:10.3390/rs17040570](https://doi.org/10.3390/rs17040570)

Certificates

Supervised Machine Learning: Regression and Classification, DeepLearning.AI and Stanford University, Coursera	Oct 2024
Applied Machine Learning in Python, University of Michigan, Coursera	Jul 2020

Languages

English: Full Professional Proficiency (TOEFL 115/120)
Persian: Native Proficiency

Achievement

Teaching Assistant Excellence Laureate Department of Computer Science, University of Calgary — awarded for outstanding teaching support (link)	2024/2025
Ranked 17th Nationwide in Iran University Entrance Exam (Konkour) Top 0.001% among all participants	May 2016